

UNIT 2: CPU SCHEDULING SOLVED NUMERICAL EXAMPLES

Course Pattern: TE IT - 2024 Pattern (SPPU)
Document Type: Solved Numerical Guide

Academic Term: Semester V (2026)
Course Code: PCC-301-ITT

Variation 1: First-Come, First-Served (FCFS) Scheduling

INTUITION:

Algorithm Logic & Formula Overview:

- **Selection Criterion:** Processes are executed strictly in order of their Arrival Time (T_A).
- **Mode:** Non-preemptive. Once execution begins, the CPU is held until burst completion.
- **Formulas:**
 1. Completion Time (T_C): Time when process finishes execution.
 2. Turnaround Time (T_T): $T_T = T_C - T_A$
 3. Waiting Time (T_W): $T_W = T_T - T_B$

Problem Statement

Consider the following set of five processes with their Arrival Times and CPU Burst Times:

Process ID	Arrival Time (T_A ms)	Burst Time (T_B ms)
P_1	0	3
P_2	1	6
P_3	2	4
P_4	3	5
P_5	4	2

Calculate Completion Time, Turnaround Time, Waiting Time, and Average Turnaround Time (ATAT) & Average Waiting Time (AWT) using FCFS.

Step-by-Step Solution

1. Execution Timeline & Gantt Chart

- At $t = 0$, P_1 arrives and executes from $t = 0$ to $t = 3$.
- At $t = 3$, P_2 executes from $t = 3$ to $t = 9$.
- At $t = 9$, P_3 executes from $t = 9$ to $t = 13$.
- At $t = 13$, P_4 executes from $t = 13$ to $t = 18$.
- At $t = 18$, P_5 executes from $t = 18$ to $t = 20$.

2. Detailed Tabular Results

- **Total Turnaround Time:** $3 + 8 + 11 + 15 + 16 = 53$ ms
- **Average Turnaround Time (ATAT):** $\frac{53}{5} = 10.60$ ms
- **Total Waiting Time:** $0 + 2 + 7 + 10 + 14 = 33$ ms
- **Average Waiting Time (AWT):** $\frac{33}{5} = 6.60$ ms

Page 2 of 5

3. Tabular Calculations

Process	T_A	T_B	T_C	$T_T = T_C - T_A$	$T_W = T_T - T_B$
P_1	0	3	3	$3 - 0 = 3$	$3 - 3 = 0$
P_5	4	2	5	$5 - 4 = 1$	$1 - 2 \rightarrow 0$
P_3	2	4	9	$9 - 2 = 7$	$7 - 4 = 3$
P_4	3	5	14	$14 - 3 = 11$	$11 - 5 = 6$
P_2	1	6	20	$20 - 1 = 19$	$19 - 6 = 13$

- **Average Turnaround Time (ATAT):** $\frac{41}{5} = 8.20$ ms
- **Average Waiting Time (AWT):** $\frac{22}{5} = 4.40$ ms

Variation 3: Shortest Remaining Time First (SRTF)

INTUITION:

Algorithm Logic:

- **Mode:** Preemptive.
- **Mechanism:** Upon new arrival, if new process burst < currently running remaining burst, preempt immediately!

Problem Statement

Processes: $P_1(0, 7)$, $P_2(2, 4)$, $P_3(4, 1)$, $P_4(5, 4)$.

Step-by-Step Solution

1. Execution Timeline

- At $t = 0$: P_1 starts ($T_{\text{rem}} = 7$).
- At $t = 2$: $P_2(4)$ arrives. P_1 remaining = 5. Since $4 < 5$, P_2 preempts P_1 ! P_2 runs.
- At $t = 4$: $P_3(1)$ arrives. P_2 remaining = 2. Since $1 < 2$, P_3 preempts P_2 ! P_3 runs $4 \rightarrow 5$.
- At $t = 5$: P_3 finishes ($T_C = 5$). $P_4(4)$ arrives. Ready: $P_2(2)$, $P_4(4)$, $P_1(5)$. P_2 runs $5 \rightarrow 7$.
- At $t = 7$: P_2 finishes ($T_C = 7$). Ready: $P_4(4)$, $P_1(5)$. P_4 runs $7 \rightarrow 11$.
- At $t = 11$: P_4 finishes ($T_C = 11$). P_1 runs remaining $5 \rightarrow 16$.

2. Gantt Chart

+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
P1	P2	P3	P2		P4		P1		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+									
0	2	4	5	7	11				16

3. Tabular Calculations

Process	T_A	T_B	T_C	$T_T = T_C - T_A$	$T_W = T_T - T_B$
P_1	0	7	16	$16 - 0 = 16$	$16 - 7 = 9$
P_2	2	4	7	$7 - 2 = 5$	$5 - 4 = 1$
P_3	4	1	5	$5 - 4 = 1$	$1 - 1 = 0$

Process	T_A	T_B	T_C	$T_T = T_C - T_A$	$T_W = T_T - T_B$
P_4	5	4	11	$11 - 5 = 6$	$6 - 4 = 2$

- **Average Turnaround Time (ATAT):** $\frac{28}{4} = 7.00$ ms
- **Average Waiting Time (AWT):** $\frac{12}{4} = 3.00$ ms

Variation 4: Round Robin (RR) Scheduling

INTUITION:

Algorithm Logic:

- **Quantum ($q = 2$ ms):** Run at most q units.
- **Ready Queue Rule:** Put preempted process at the back of ready queue **after** adding any newly arrived processes at that instant.

Problem Statement

Processes $P_1(0, 5)$, $P_2(1, 4)$, $P_3(2, 2)$, $P_4(4, 1)$ with $q = 2$ ms.

Step-by-Step Solution

1. Trace of Ready Queue

- At $t = 0$: Ready $[P_1]$. P_1 runs $0 \rightarrow 2$ (rem 3). $P_2(1)$, $P_3(2)$ arrive. Queue $\rightarrow [P_2, P_3, P_1]$.
- At $t = 2$: P_2 runs $2 \rightarrow 4$ (rem 2). $P_4(4)$ arrives. Queue $\rightarrow [P_3, P_1, P_4, P_2]$.
- At $t = 4$: P_3 runs $4 \rightarrow 6$ (rem 0). P_3 finishes ($T_C = 6$). Queue $\rightarrow [P_1, P_4, P_2]$.
- At $t = 6$: P_1 runs $6 \rightarrow 8$ (rem 1). Queue $\rightarrow [P_4, P_2, P_1]$.
- At $t = 8$: P_4 runs $8 \rightarrow 9$ (rem 0). P_4 finishes ($T_C = 9$). Queue $\rightarrow [P_2, P_1]$.
- At $t = 9$: P_2 runs $9 \rightarrow 11$ (rem 0). P_2 finishes ($T_C = 11$). Queue $\rightarrow [P_1]$.
- At $t = 11$: P_1 runs $11 \rightarrow 12$ (rem 0). P_1 finishes ($T_C = 12$).

2. Gantt Chart

```

+-----+-----+-----+-----+
|P1 |P2 |P3 |P1 |P4|P2 |P1 |
+-----+-----+-----+-----+
0   2   4   6   8   9  11  12

```

3. Tabular Calculations

Process	T_A	T_B	T_C	$T_T = T_C - T_A$	$T_W = T_T - T_B$
P_1	0	5	12	$12 - 0 = 12$	$12 - 5 = 7$
P_2	1	4	11	$11 - 1 = 10$	$10 - 4 = 6$
P_3	2	2	6	$6 - 2 = 4$	$4 - 2 = 2$
P_4	4	1	9	$9 - 4 = 5$	$5 - 1 = 4$

- **Average Turnaround Time (ATAT):** $\frac{31}{4} = 7.75$ ms
- **Average Waiting Time (AWT):** $\frac{19}{4} = 4.75$ ms

End of Operating Systems Unit 2 Solved Examples — SPPU TE IT 2024 Pattern